Lecture 25 (Complimentary Slackness and Dual Simplex Method)

April 29, 2026

Today we will talk about complimentary slackness and briefly describe the dual simplex method (without proof).

Theorem (Complementary Slackness)

Let $\vec{x}$ and $\vec{p}$ be feasible solutions to a primal LP and its corresponding dual LP. The vectors $\vec{x}$ and $\vec{p}$ are optimal in their respective LPs if and only if the following two conditions are met:

1. $p_i\left(\vec{a}_i\vec{x} - b_i\right) = 0 \ \forall i$
2. $\left(c_j - \vec{p}\cdot A_j\right)x_j = 0 \ \forall j$

Where $\vec{a}_i$ is the ith row of matrix A in the primal LP (as a row vector) and A_j is the jth column of matrix A. We call this pair of conditions, **Complimentary Slackness**.

Proof

Recall from our proof of weak duality we can set

$$u_i = p_i\left(\vec{a}_i\vec{x} - b_i\right)$$

$$v_j = \left(c_j - \vec{p}\cdot A_j\right)x_j$$

And we showed that if $\vec{x}, \vec{p}$ are feasible in their respective LPs that

$$u_i, v_j \geq 0 \ \forall i, j$$

We also showed that

$$\vec{c}\cdot\vec{x} - \vec{p}\cdot\vec{b} = \sum_i u_i + \sum_j v_j$$

where $\vec{c}\cdot\vec{x}$ and $\vec{p}\cdot\vec{b}$ are the objectives of the primal and dual LPs respectively. If $u_i, v_j = 0$ for all i, j then we see

$$\vec{c}\cdot\vec{x} - \vec{p}\cdot\vec{b} = \sum_i u_i + \sum_j v_j = 0$$

Now, suppose $\vec{x}, \vec{p}$ are optimal in their respective LPs. Then by strong duality, we see

$$\vec{c}\cdot\vec{x} = \vec{p}\cdot\vec{b}$$

So

$$0 = \vec{c}\cdot\vec{x} - \vec{p}\cdot\vec{b} = \sum_i u_i + \sum_j v_j$$

but as each $u_i, v_j \geq 0$ for all i, j this implies that $u_i, v_j = 0$ for all i, j. Now, suppose $u_i, v_j = 0$ for all i, j. Then

$$0 = \sum_i u_i + \sum_j v_j = \vec{c}\cdot\vec{x} - \vec{p}\cdot\vec{b}$$

So

$$\vec{c}\cdot\vec{x} = \vec{p}\cdot\vec{b}$$

As both objectives are equal for some feasible solutions $\vec{x}, \vec{p}$, then by weak duality we see both $\vec{x}$ and $\vec{p}$ are optimal in their respective LPs.

Notes

If the primal solution is in standard form, then

$$A\vec{x} = \vec{b}$$

form the constraints. This implies each individual constraint is of the form

$$\vec{a}_i\vec{x} = b_i$$

so we see from complimentary slackness that

$$p_i\left(\vec{a}_i\vec{x} - b_i\right) = 0 \ \forall i$$

for free. This means we can check is a solution is optimal by checking $\left(c_j - \vec{p}\cdot A_j\right)x_j = 0 \ \forall j$ only.

Ex 1)

Consider the following LP

$$\text{minimize } 13x_1 + 10x_2 + 6x_3$$

such that

$$5x_1 + x_2 + 3x_3 = 8$$

$$3x_1 + x_2 + 0x_3 = 3$$

$$x_1, x_2, x_3 \geq 0$$

This yields the dual LP

$$\text{maximize } 8p_1 + 3p_2$$

such that

$$5p_1 + 3p_2 \leq 13$$

$$p_1 + p_2 \leq 10$$

$$3p_1 + 0p_2 \leq 6$$

$$p_1, p_2 \in \mathbb{R}$$

Suppose we assume a feasible solution exists for the primal LP. Then (1) of complementary slackness is satisfied as

$$p_1\left(5x_1 + x_2 + 3x_3 - 8\right) = p_1\left(0\right) = 0$$

$$p_2\left(3x_1 + x_2 - 3\right) = p_2\left(0\right) = 0$$

for any feasible solution to the primal LP. Now, we can check (2) of complementary slackness. We have three possible conditions to check:

$$\left(13 - \left(5p_1 + 3p_2\right)\right)x_1$$

$$\left(10 - \left(p_1 + p_2\right)\right)x_2$$

$$\left(6 - 3p_1\right)x_3$$

If we suppose there exists an optimal primal solution we will only have at most two active variables in the optimal solution. Suppose we have x_1, x_3 active and $x_2 = 0$ inactive. Then we see

$$(10 - (p_1 + p_2))\, x_2 = (10 - (p_1 + p_2))\, (0) = 0$$

so we only need to see if there exist p_1, p_2 to make the remaining constraints equal to zero. Since we are assuming x_1, x_3 are active, we have $x_1, x_3 \geq 0$. If we assume the LP is not degenerate, we can further assume $x_1, x_3 > 0$, thus any optimal $\vec{p}$ must satisfy

$$(13 - (5p_1 + 3p_2))\, x_1 = (13 - 5p_1 - 3p_2)\, x_1 = 0$$

$$(6 - 3p_1)\, x_3 = 0$$

so

$$(13 - 5p_1 - 3p_2) = 0$$

$$(6 - 3p_1) = 0$$

And as the linear system has a solution, we see

$$p_1 = 2, p_2 = 1$$

satisfies the linear system (and thus complementary slackness). This tells us the dual cost

$$8\,(2) + 3\,(1) = 19$$

is optimal to the dual LP. By strong duality, this is also the optimal value to the primal LP.

Ex 2)

We can generalize **Ex 1)**. Suppose x_j is a basic variable with a non-degenerate optimal BFS to the primal problem in standard form. Then

$$(c_j - \vec{p} \cdot A_j)\, x_j = 0 \implies \vec{p} \cdot A_j = c_j$$

for all j such that x_j is active. If the columns of A are linearly independent, we can say

$$\vec{p}^T = \vec{c}_B^T B^{-1}$$

With this result, we can justify the dual simplex method. We will not prove this method explicitly, but the proof is similar to the proof for the standard simplex method thanks to the above result.

Dual Simplex Method

Recall the simplex tableau:

$$\begin{array}{c|c} -\vec{c}_B^T B^{-1}\vec{b} & \vec{c} - \vec{c}_B^T B^{-1} A \\ - & - \\ B^{-1}\vec{b} & B^{-1}A \end{array}$$

(Here I added some $-$ and $|$ symbols to separate the blocks of the tableau. This is only in the typesetting). We often write this in terms of the reduced cost:

$$\begin{array}{c|c} -\vec{c}_B^T B^{-1}\vec{b} & \vec{\bar{c}} \\ - & - \\ B^{-1}\vec{b} & B^{-1}A \end{array}$$

The key idea of the **dual simplex** method is the work with the transpose of A when doing operations. The steps are given below. I encourage you to go through the proof of the simplex tableau method and see if you can prove the dual simplex using a similar method. At the least, you might find it helpful to compare the steps of this algorithm to those in the standard simplex tableau.

1. Start with a basis matrix B such that all reduced costs are non-negative ($B^{-1}\vec{b}$ may have negative entries).
2. Check the 0th column of the tableau. If all of the entries are non-negative, we have an optimal BFS and terminate. Otherwise, choose l such that $x_{B(l)} < 0$.
3. Consider the lth row of the tableau with elements $x_{B(l)}, v_1, \ldots, v_n$. If $v_i \geq 0$ for all i, then the optimal cost is $+\infty$ and the algorithm terminates.
4. For all i where $v_i < 0$, compute $\frac{\bar{c}_i}{|v_i|}$ and let j be the index of a column with the smallest ratio. Column $A_{B(l)}$ exits the basis and A_j enters.
5. Add to each row of the tableau a multiple of the lth row so v_j becomes 1 and all other entries go to 0.

Notes

We can still use lexicographic ordering and Bland's rule for pivoting. We will not prove that this is the case. The book has details (but I encourage you to try proving this result on your own if you are interested).

Why Use Dual Simplex?

1) Sometimes a BFS is easier to find for the dual LP than for the primal.

2) If you have the same objective and A matrix in the primal for a collection of related LPs, but the values of $\vec{b}$ change, then the dual simplex method will allow you to initialize with a simple dual feasible solution. The dual feasibility is unaffected by $\vec{b}$, so you can save significant compute time on the initialization steps.